

Joseph Zou

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EDUCATION

Purdue University

West Lafayette, IN

Bachelor of Science in Data Science, GPA: 3.91/4.0

Aug. 2023 – May 2026

Relevant Coursework: Artificial Intelligence, Data Mining & Machine Learning, Statistical Programming & Data Management, Data Visualization, Data Structures & Algorithms, Python Programming, Statistics for Data Science

ASA DataFest: Best Statistical Analysis Award, Purdue University (Spring 2024)

EXPERIENCE

Data Science Researcher @ John Deere

Aug 2025 – Present

- Designed and trained tree-based machine learning models (Decision Trees, Random Forests, XGBoost) for 3-month part-level demand forecasting across multi-year time series data
- Implemented recursive forecasting to generate realistic, sequential predictions without data leakage
- Engineered lag-based and temporal features to capture short-term demand patterns across 67-month time series
- Evaluated model performance using RMSE to guide improvements in forecast accuracy and business relevance

Data Science Intern @ BASF

Aug 2024 – May 2025

- Analyzed 10 years of California miticide application data to identify seasonal and regional usage patterns
- Estimated reapplication probabilities conditioned on spring active ingredient choice
- Conducted cost-benefit simulations to compare long-term outcomes of treatment strategies
- Performed geospatial county-level segmentation to inform region-specific product recommendations

Machine Learning Research Assistant @ CourseKata Project

March 2024 – June 2024

- Built linear regression models on 450K student records to identify key factors influencing academic outcomes
- Applied correlation analysis and k-means clustering to video engagement logs, finding rewatchers were 72% more likely to complete assignments
- Engineered data pipelines in R (tidyverse) and created ggplot2 visualizations to support feature analysis
- Delivered insights at ASA DataFest through reproducible workflows, engaging academic and industry audiences

PROJECTS

NBA Prop Prediction | *Python, Pandas, scikit-learn*

December 2025 – Present

- Built an end-to-end pipeline to predict NBA player points using historical performance and game context
- Engineered rolling and opponent-specific features to capture recent form, minutes trends, and matchup effects
- Trained and evaluated linear regression and gradient-boosted tree models using RMSE and MAE on held-out data
- Developed model interpretability and evaluation visualizations, including feature importance and prediction calibration

Body Image & Gym Habits ML Analysis | *Python, Pandas, NumPy, Matplotlib*

May 2025 – August 2025

- Built an end-to-end machine learning pipeline on behavioral survey data, from preprocessing to evaluation
- Trained and compared logistic regression, decision tree, and random forest models to predict body dissatisfaction risk based on behavioral and demographic survey variables
- Evaluated model performance using ROC-AUC and recall across linear and tree-based classifiers
- Applied K-means and hierarchical clustering to uncover gym-goer archetypes and quantify large disparities in body dissatisfaction risk, including one cluster with ~68% at-risk prevalence

Swimlytics | *R, k-NN, t-test, ggplot2, tidyverse*

May 2024 – July 2024

- Investigated trends in youth swimming to explore fairness and performance equity
- Cleaned and structured raw competition results for EDA and hypothesis testing
- Ran two-sample t-tests to assess statistical significance of gender performance gaps
- Built a k-NN model with 93% accuracy to predict swimmer gender using age and race time features

TECHNICAL SKILLS

Languages: Python, R, SAS, Java, C/C++, SQL **Libraries:** pandas, NumPy, scikit-learn, matplotlib, ggplot2, NLTK